

RANPilotAI

AI-Powered RAN Performance Optimization Platform

Graduation Project ■ 2026

Tasneem Amein ■ Ibrahim Samy ■ Khaled Mogahed
Mahmoud Ashraf ■ Mohab Tarek

Supervisor: **Dr. Salma Sobhy**

Agenda

- 1 Introduction, Architecture & KPI Degradation
- 2 KPI Forecasting Engine
- 3 Optimization Action Analyzer
- 4 Telecom RAG Assistant
- 5 Software Testing & Experimental Results

Tasneem Amein

Problem Statement ▪ Proposed Solution ▪ Overall Architecture ▪
KPI Degradation Analyzer

Problem Statement

[Fill in your content here]

Project Motivation & Objectives

[Fill in your content here]

Proposed Solution Overview

[Fill in your content here]

Overall System Architecture

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Screenshot not yet added

screenshots/tasneem/hub_landing.png

KPI Degradation Analyzer — Module Overview

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KPI Categories Monitored

[Fill in your content here]

Detection Pipeline & Root Cause Analysis

[Fill in your content here]

KPI Degradation Analyzer — Interface

Screenshot not yet added

screenshots/tasneem/kpi_degradation_ui.png

KPI Degradation — Sample Results

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Ibrahim Samy

KPI Forecasting Engine ▪ Models ▪ Feature Engineering ▪
Accuracy & Results

Forecasting Module Overview & Motivation

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Forecasting Approach — XGBoost & Holt-Winters

[Fill in your content here]

Feature Engineering & Seasonality Decomposition

[Fill in your content here]

Forecasting Module — Interface

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screenshots/ibrahim/forecasting_ui.png

Model Evaluation & Accuracy Metrics

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7-Day KPI Forecast — Output

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XGBoost Feature Importance

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Khaled Mogahed

Optimization Action Analyzer ▪ Configuration Tracking ▪ Impact Quantification

Action Analyzer — Module Overview

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Configuration Tracking with Dolt

[Fill in your content here]

Action Analyzer — Interface

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Action Timeline View

Screenshot not yet added

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Before / After KPI Impact Analysis

Before Configuration Change

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After Configuration Change

Screenshot not yet added

screenshots/khaled/after_change.png

KPI Delta Bar Chart

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Mahmoud Ashraf

Telecom RAG Assistant ▪ Pipeline ▪ Interface ▪
Anti-Hallucination Strategy

RAG Assistant — Overview & Motivation

The Problem

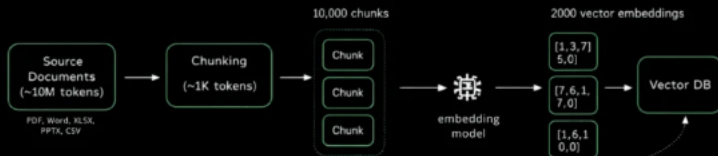
- Engineers waste hours manually searching 3GPP PDF specs
- Keyword search fails when wording differs from the spec
- General LLMs **hallucinate** protocol details
- No tool provides citation-backed, spec-grounded answers

Our Solution

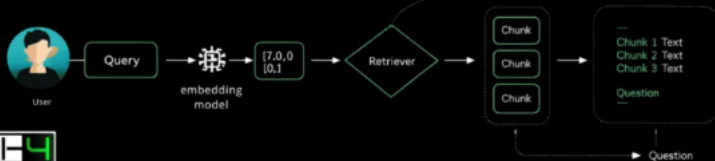
- **Engineering-grade RAG** purpose-built for 3GPP
- Fully **local**: embeddings, vector store, re-ranking
- Every answer cites: TS number, Release, Section, page
- Integrated into RANPilotAI — no tool switching

RAG Pipeline Overview

1. Knowledge-base construction (Ingestion pipeline):



2. Retrieval Pipeline



eph4.ai

Pipeline — Key Engineering Decisions

Ingestion & Chunking

- **Section-aware** PDF parsing (PyMuPDF) splits on 3GPP section boundaries — formulas and ASN.1 never cut mid-way
- Target chunk: 200–350 tokens (max 500)
- 8 chunk types: section, event_def, ASN.1, table, figure, annex, history, formula

Retrieval & Generation

- **BGE-large-en-v1.5** embedder → local **Qdrant** vector store
- Top-20 retrieved → cross-encoder reranker → top-5 sent to LLM
- **Anti-hallucination prompt:** model must say “*cannot find in specifications*” rather than guess
- Metadata filter: restrict to specific TS / Release

RAG Assistant — Main Interface

RANPilotAI
Network Optimization Toolkit

TOOLS

3GPP RAG Assistant

4G(LTE)/5G(NR) KPI Degradation Analyzer

4G(LTE)/5G(NR) KPI Forecaster

Optimization Action Analyzer

3GPP RAG Assistant

Select a tool above to load its interface.

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3GPP RAG Assistant

SPECIFICATION FILTER

Narrow the search to a specific 3GPP document.

All

SESSION

3GPP RAG Assistant

Ask engineering questions — answers grounded directly in 3GPP specifications

3GPP Specs Semantic Search Cross-Encoder Reranking LLM Generation Cited Answers

Start a conversation

Ask any question about LTE/NR procedures, measurements, events, or 3GPP protocols. Answers are cited directly from the loaded specifications.

Try asking:

Explain Event A3 in LTE

What is the RRC connection re-establishment procedure?

How does PDCP integrity protection work in 5G NR?

What triggers a handover failure in LTE?

Explain the difference between SRB1 and SRB2

What is Time-To-Trigger (TTT) in measurement events?

e.g., Explain Event A3 in LTE, or ask any 3GPP question...

Live Query — Pipeline Execution Steps

The screenshot displays a user interface for a live query. At the top, a grey input bar contains the text "what is PDCP?". Below this, a yellow icon of a person with a speech bubble is followed by a list of five execution steps, each with a small icon and a progress bar:

- Embedding query...
- Searching vector database...
- Reranking retrieved chunks...
- Building prompt context...
- Generating answer via LLM...

At the bottom, a light blue rounded rectangle contains the placeholder text "e.g., Explain Event A3 in LTE, or ask any 3GPP question..." and a right-pointing arrow.

Retrieved Chunks & Source Citations

RANPilotAI
Network Optimization Toolkit

TOOLS

3GPP RAG Assistant

4G(LTE)/5G(NR) KPI Degradation Analyzer

4G(LTE)/5G(NR) KPI Forecaster

Optimization Action Analyzer

3GPP RAG Assistant

Select a tool above to load its interface.

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3GPP RAG Assistant

SPECIFICATION FILTER

Narrow the search to a specific 3GPP document.

All

SESSION

1 Citation(s)

what is PDCP?

PDCP (Packet Data Convergence Protocol)

PDCP is the layer in the 5G NR (and LTE) protocol stack that sits between the RLC and the higher-layer protocols (RRC, NAS, IP). Its main functions are:

Function	Purpose
Header compression	Reduces the size of IP/UDP/TCP headers (e.g., via RLC header compression or Ethernet header compression).
Security	Provides confidentiality, integrity, and authentication for user data and control messages.
Sequence numbering	Adds a sequence number to each PDCP SDU to enable re-ordering and loss detection.
Duplicate detection	Detects and discards duplicate PDCP SDUs that may arrive because of retransmissions or packet duplication.
Re-ordering	Re-orders out-of-order PDCP SDUs before passing them to the upper layer.
Status reporting	For AM RLC bearers, the PDCP can send status reports to the RLC to inform about received SDUs.
Split-bearer support	Allows a single PDCP entity to be associated with multiple RLC entities (e.g., for carrier aggregation or IAB).

e.g., Explain Event A3 in LTE, or ask any 3GPP question...

Answer Panel & Backend Response

Citation-Backed Answer Panel

The screenshot displays the RANPilot AI interface. On the left, a sidebar contains a 'TOOLS' section with a dropdown menu set to '3GPP RAN Assistant'. Below this, a 'SPECIFICATION FILTER' section shows a search for 'A3'. The main panel features a search bar with the query 'Explain A3 Event in LTE?'. Below the search bar, the title 'A3 Event in LTE (NR)' is followed by a definition: 'Definition (from the specification):'. A bulleted list explains the event: 'Event A3: Neighbour becomes amount of offset better than PCell/PSCell [11.1.4.4]'. It further states: 'In other words, the UE declares Event A3 when the measurement of a neighboring cell (e.g., RSRP, RSRQ, or SRSR) exceeds the measurement of the serving cell (PCell or PSCell) by a configured offset.' Below this, the 'Trigger Condition (exact wording from the spec)' is quoted: '“Neighbour becomes amount of offset better than PCell/PSCell”'. A note explains: 'This means the UE compares the measured quantity of the neighbour cell with the measured quantity of the serving cell and the neighbour must be greater by the configured offset value.' A table titled 'Parameters included in the EventA3 trigger configuration' lists 'A3-Offset' with its type and meaning. At the bottom, a text input field contains the query 'e.g., Explain Event A3 in LTE, or ask any 3GPP question...' with a right arrow button.

TOOLS

3GPP RAN Assistant

4G/LTE/5G/NR KPI Degradation Analyzer

4G/LTE/5G/NR KPI Forecaster

Optimization Action Analyzer

3GPP RAN Assistant

Select a tool above to load its interface.

Module: Topic: (1) Configuration 3GPP

3GPP RAN Assistant

SPECIFICATION FILTER

Narrow the search to a specific 3GPP document

A3

SESSION

Explain A3 Event in LTE?

A3 Event in LTE (NR)

Definition (from the specification):

- Event A3: Neighbour becomes amount of offset better than PCell/PSCell [11.1.4.4]
- In other words, the UE declares Event A3 when the measurement of a neighboring cell (e.g., RSRP, RSRQ, or SRSR) exceeds the measurement of the serving cell (PCell or PSCell) by a configured offset.

Trigger Condition (exact wording from the spec)

“Neighbour becomes amount of offset better than PCell/PSCell”

This means the UE compares the measured quantity of the neighbour cell with the measured quantity of the serving cell and the neighbour must be greater by the configured offset value.

Parameters included in the EventA3 trigger configuration

Parameter	Type	Meaning (as defined in TS 38.331)
A3-Offset	Offset measurement	Offset value(s) to be used in the NR conditional reconfiguration triggering condition for configured A3. The actual offset is [offset value] + [offset]

e.g., Explain Event A3 in LTE, or ask any 3GPP question...

Backend Pipeline Stages

The screenshot displays the RANPilot AI interface. At the top, a search bar contains the query 'what is PDCP?'. Below the search bar, a list of stages is shown: 'Embedding query...', 'Searching vector database...', 'Ranking retrieved chunks...', 'Building prompt context...', and 'Generating answer via LLM...'. At the bottom, a text input field contains the query 'e.g., Explain Event A3 in LTE, or ask any 3GPP question...' with a right arrow button.

what is PDCP?

- Embedding query...
- Searching vector database...
- Ranking retrieved chunks...
- Building prompt context...
- Generating answer via LLM...

e.g., Explain Event A3 in LTE, or ask any 3GPP question...

Mohab Tarek

Software Testing ▪ Real-Case Experimental Results ▪ Conclusion
▪ Future Work

Testing Strategy & Test Coverage

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Unit & Integration Testing

[Fill in your content here]

Real-Case Dataset Description

[Fill in your content here]

KPI Degradation — Experimental Results

Screenshot not yet added

screenshots/mohab/results_degradation.png

Forecasting — Experimental Results

Screenshot not yet added

screenshots/mohab/results_forecasting.png

Action Analyzer — Experimental Results

Configuration Change Event

Screenshot not yet added

screenshots/mohab/results_action_before.png

KPI Impact Measured

Screenshot not yet added

screenshots/mohab/results_action_after.png

RAG Assistant — Experimental Results

Screenshot not yet added

screenshots/mohab/results_rag.png

Conclusion

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[Fill in your content here]

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